Practice workbook · Define pathogen, classify diseases and name the organism type causing each

The term pathogen, communicable versus non-communicable conditions, and matching named diseases to virus, bacterium, protozoan or fungus.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-PATHOGEN-AND-NAME-CAUSE

1. Question 3 · part a · 2 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

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.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

1. Question 3 · part a · 2 marks · Working and answer

Working and answer

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2. Question 4 · part b · 1 mark

June 2019 Paper 2

Further practice · Attempt part b only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

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number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

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.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

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2. Question 4 · part b · 1 mark · Working and answer

Working and answer

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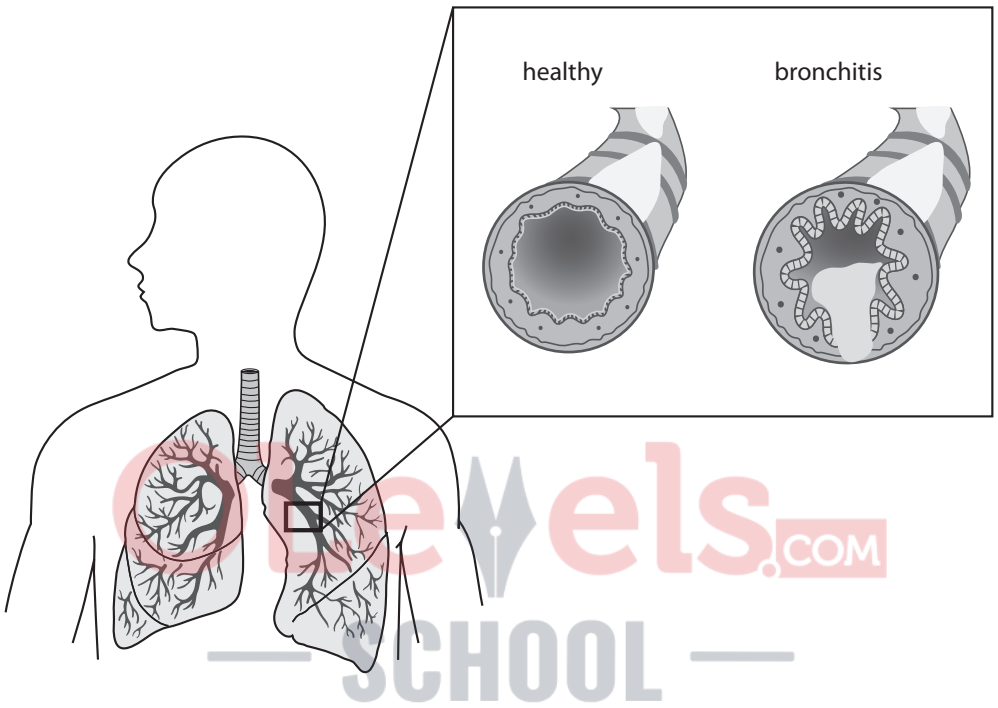
3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

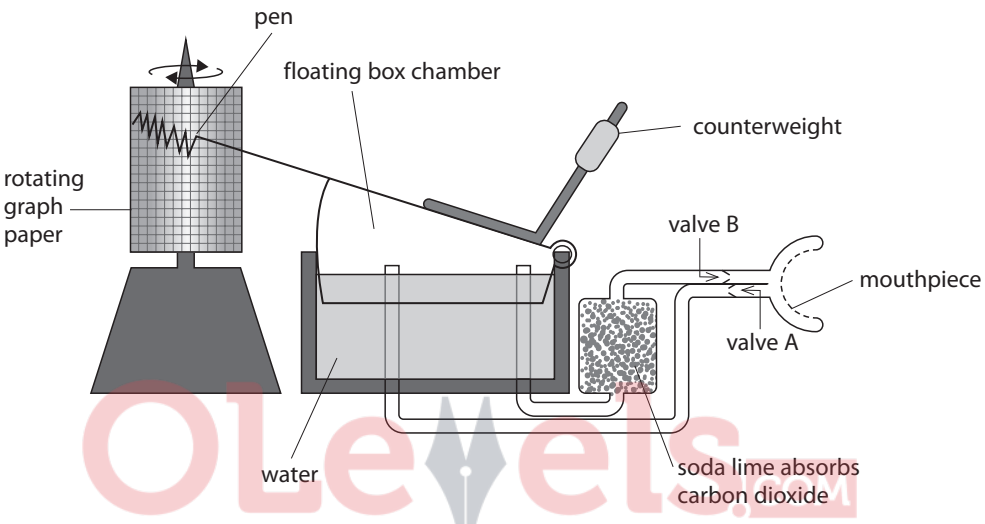
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

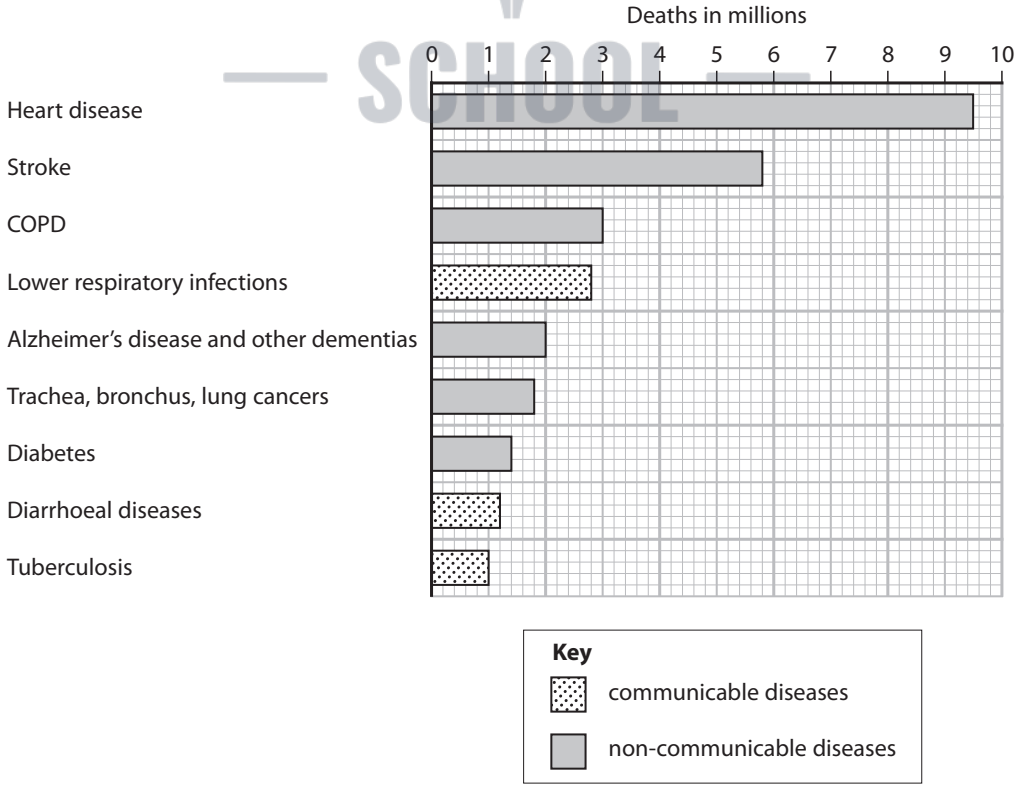
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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P 5 9 9 0 9 A 0 1 6 2 8

3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

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(i) More deaths are caused by non-communicable diseases than by communicable diseases.

Suggest what is meant by the term **non-communicable disease**.

(1)

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(ii) In 2000, the number of deaths from heart disease was 7.0 million.

Calculate the percentage change in the number of deaths from heart disease from 2000 to 2016.

(3)

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percentage change = ...%

(iii) Suggest a reason for the change in the number of deaths from heart disease from 2000 to 2016.

(1)

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(Total for Question 6 = 13 marks)

3. Question 6 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 7 · part a.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part a.i only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease?

(1)

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics.

(2)

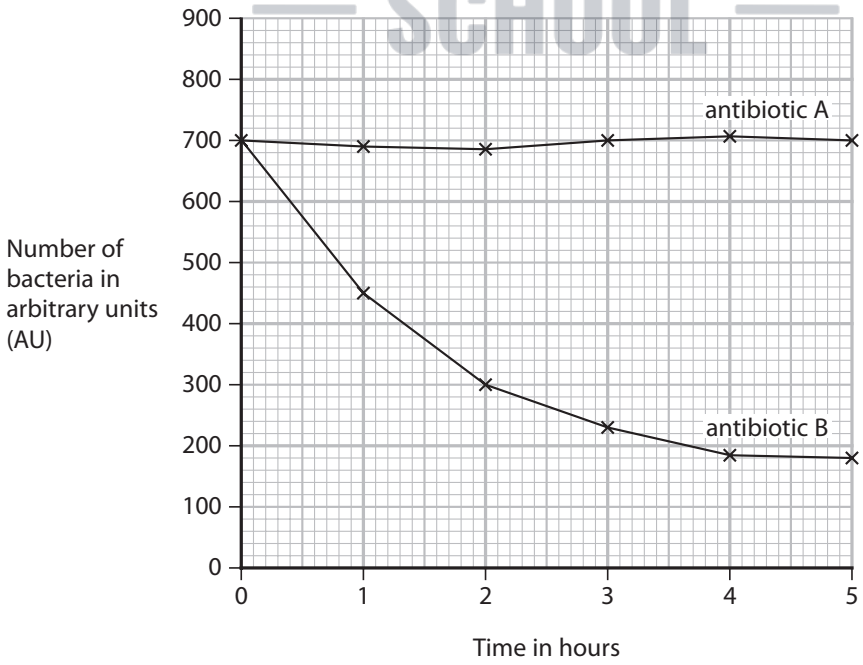
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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4. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 8 · part a.ii · 1 mark

June 2023 Paper 1R

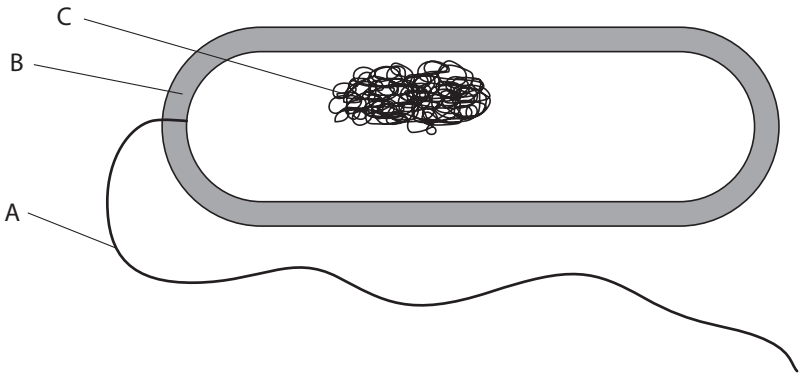
Further practice · Attempt part a.ii only.

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8 The diagram shows a bacterial cell.



(a) (i) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

(1)

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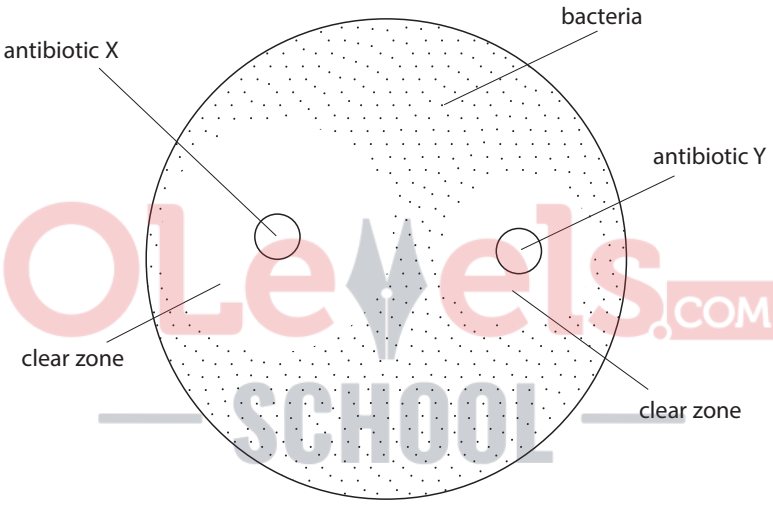
5. Question 8 · part a.ii · 1 mark

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days



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(i) Describe what is meant by the term **antibiotic**.

(2)

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P 7 2 5 8 6 A 0 1 8 2 0

5. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 3 · part a · 1 mark

November 2023 Paper 2

Further practice · Attempt part a only.

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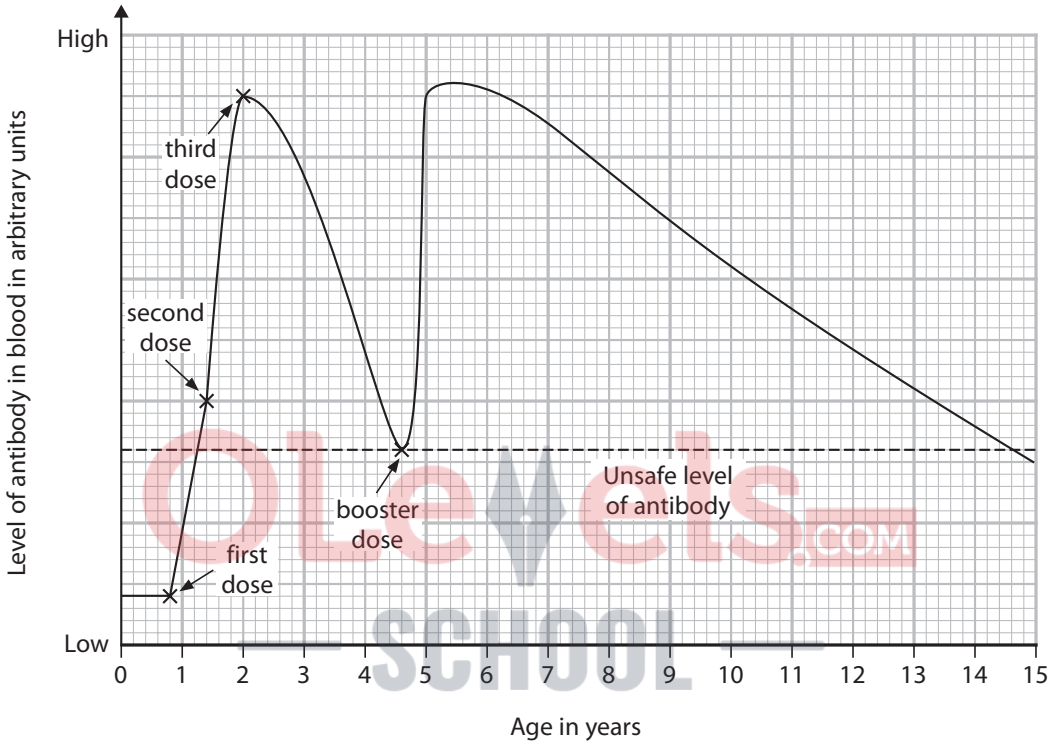
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3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

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6. Question 3 · part a · 1 mark · Working and answer

Working and answer

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7. Question 4 · part a · 1 mark

June 2025 Paper 1

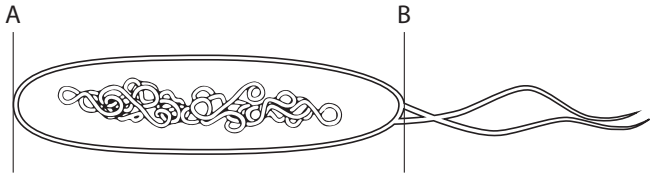
Further practice · Attempt part a only.

4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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7. Question 4 · part a · 1 mark · Working and answer

Working and answer

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8. Question 3 · part a · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

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2 .....

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3 .....

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(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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8. Question 3 · part a · 1 mark · Working and answer

Working and answer

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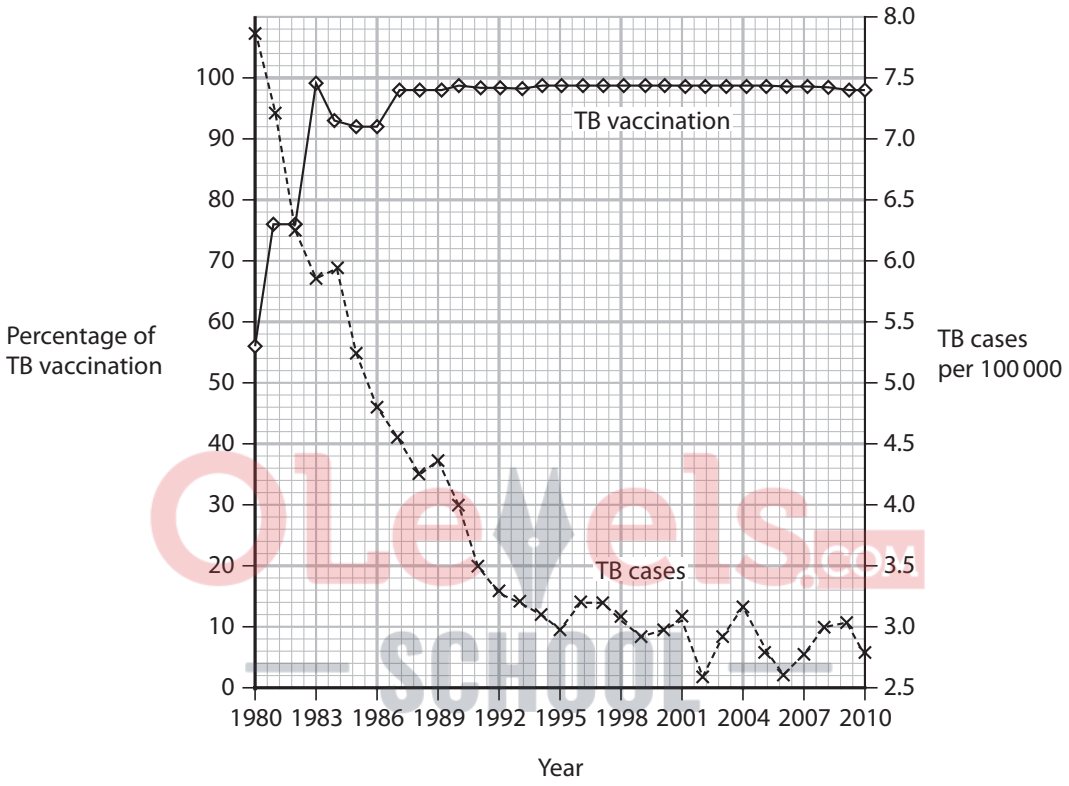
9. Question 4 · part a.i · 1 mark

June 2021 Paper 1

Further practice · Attempt part a.i only.

4 The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.



(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

(1)

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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9. Question 4 · part a.i · 1 mark

June 2021 Paper 1

Further practice · Attempt part a.i only.

(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

(4)

(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

(3)

number of cases = ...

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9. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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10. Question 7 · part a · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a only.

7 Cholera is a disease that causes chronic water loss from the body.

(a) Name the type of microorganism that causes cholera.

(1)

(b) Describe how cholera is transmitted from one person to another.

(2)

(c) Explain how a person with cholera can be treated for the disease.

(4)

(Total for Question 7 = 7 marks)

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10. Question 7 · part a · 1 mark · Working and answer

Working and answer

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11. Question 8 · part a.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.i only.

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8 (a) Cholera is a disease that kills approximately 95 000 people each year worldwide.

(i) Name the type of pathogen that causes cholera.

(1)

(ii) State how cholera is transmitted from one person to another.

(1)

(b) Describe a method used to treat a person with cholera.

(3)

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(Total for Question 8 = 5 marks)

Working and answer

12. Question 1 · part a · 3 marks

June 2024 Paper 2

Further practice · Attempt part a only.

1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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12. Question 1 · part a · 3 marks · Working and answer

Working and answer

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13. Question 8 · part b.i · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.i only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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13. Question 8 · part b.i · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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(b) In 2010, AIDS-related illnesses caused the death of 1.4 million people.

In 2021, AIDS-related illnesses caused the death of 650 000 people.

(i) Name the microorganism that causes AIDS.

(1)

.....

(ii) Calculate the percentage decrease in the number of people dying from AIDS-related illnesses between 2010 and 2020.

(2)

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percentage decrease = ... %

(Total for Question 8 = 10 marks)

Working and answer

14. Question 1 · part a · 6 marks

November 2020 Paper 2

Further practice · Attempt part a only.

1 (a) Different types of organisms cause diseases.

Diseases are transmitted in different ways.

The table lists three diseases.

Complete the table by giving the type of organism that causes each disease and the method of transmission for each disease.

(6)

Name of disease	Type of organism	Method of transmission
malaria		
poliomyelitis		
typhoid		

(b) Explain why it is not possible to treat athlete's foot using antibiotics.

(2)

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(Total for Question 1 = 8 marks)

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14. Question 1 · part a · 6 marks · Working and answer

Working and answer

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